

H. Reza Mohammadiha

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Education



Sharif University of Technology
Iran

Tehran,

Master of Science in Industrial Engineering,

2020-2023

GPA: 3.46/4 or 17.05/20

Thesis: Investigating the Application of Game Theory in Pricing and Market Analysis (2023)

Supervisor: Dr. Shokrane K. Moghaddam



Sharif University of Technology
Iran

Tehran,

Bachelor of Science in Industrial Engineering,

2016-2020

GPA: 3.93/4 or 18.06/20

Research Interests

#Applied Optimization

#Computer Simulation

#Game Theory

#Queueing Theory

#Supply Chain Management

#Serious Games

#Agent Based Modeling

#Reinforcement Learning

#Dynamic Programming

Computer Skills

Digital Literacy

Data Analytics

Machine Learnings

Simulation Tools

Product Development

Web Development

Management Tools

LaTeX editor

Microsoft Office – Google Cloud – Github – LLM/LRM chatbots

Excel – Python – R – SQL – PowerBI – Tableau – Metabase

Pandas – Numpy – Scikit-Learn – Gymnasium

AnyLogic – Anylogistix – Arena

Figma – Docker – Postman

HTML – CSS – JS

Trello – Jira

Overleaf

Languages

English

Professional (IELTS: 6/9 - Duolingo: 125/160)

Turkish

Familiar

Persian

Native

Experiences



Idekavan Group
Product Manager

5 Months

Supporting the growth of small and medium-sized businesses by de-monopolizing technological power and helping these businesses to be visible and provide better services to the public. Small and medium-sized businesses face many challenges in experiencing activity in the digital space; My role was building and managing a transaction-based product as a new financial tool for small business to gather all available credits in the market of Iran.



Saman Insurance
Associate BI Developer

10 Months

Saman insurance company provide diverse and creative insurance services to individuals and legal entities with flexible payment options. My role is exploring innovative data frames and building insightful dashboard to analyze the profitability of different fields.



Iran Credit Scoring
Associate Product Manager

10 Months

Iran Credit Scoring Company develops explainable Artificial Intelligences (XAI) for the FinTech Ecosystem of Iran. My role was to model technical data pipelines and business workflow to ensure the reliability of system and clarity of machine learning algorithms.



Solico Group

30 Months

Game Studio Director

6 Months

Strategic Planning Team Member

18 Months

Organization Designer

6 Months

Solico Group is a multinational food company. It produces, trades, distributes, and sells food in more than 10 countries. My role was designing organizational charts for different departments and then modeling framework of jobs. After a while I was assigned to manage a talent pool to hunt and help young talent to know the business and find their suitable departments. At the end, I tried to establish a game studio to integrate gamification project all over the company.



Gamein 2021
Lead Game Designer

12 Months

Gamein is an academic competition in which participants face business challenges in a simulated market environment. I was like one of founding fathers of this international and academic serious game event in the field of supply chain and business management between Industrial Engineers. We designed, developed and played our game successfully between 964 participants.

Academic Projects



Industrial Engineering Department

Teaching Assistant

36 Months

- Product Development and Planning
- Simulation Languages
- Discrete Event Simulation

Dr. Shokraneh K. Moghaddam
Dr. Erfan Hassannayebi
Dr. Nafiseh Sedghi

Spring 2022
Spring 2021
Spring 2020

IBM HR analytics employees case study

2022

@ Sharif (Masters)

Data driven examining of job burnout triggers

Leveraging IBM's HR analytics dataset, this academic case study applied data science techniques to examine the multifaceted triggers of employee burnout. Through statistical analysis and predictive modeling, it identified key factors—such as overtime, job role, and employee satisfaction—that contribute to attrition, providing actionable insights for improving workforce retention strategies.

Market Simulation of Gamein 2021

2021

@ Sharif (Gamein)

Supply chain pricing mechanism design and balancing

This project involved designing and simulating a complex supply chain market for a business simulation game. The core challenge was to develop a robust pricing and trading mechanism that would balance the in-game economy, ensuring it was both competitive and sustainable for all players participating in the simulated market environment.

Hospitals' Beds Allocations for Covid-19 Patients

2020

@ Sharif (bachelor)

Game Theoretical Approach with Auction and Matching Model

Addressing a critical pandemic-era challenge, this research proposed a resource allocation model for hospital beds using a combination of game theory, auction mechanisms, and matching algorithms. The aim was to create a fair and efficient system to optimally distribute scarce medical resources among hospitals and patients, maximizing overall welfare during a crisis.

Serious Game Design for Learning Supply Chains Contracts

2019

@ Internship

At Iran's National Elites Foundation

At Iran's National Elites Foundation, this project involved designing a "serious game"—a game with an educational purpose rather than pure entertainment. The objective was to create an interactive simulation to teach students and professionals the complexities and strategic implications of different supply chain contracts in an engaging and practical manner.

Professional Projects

Financial Tool Design and Development

2026

@ Idekavan Group

Creating a payment hub for small and medium size businesses

This project was on developing a financial product as a hub for several credit payment gateways – such as Digipay, Tara, Doctorpay, - beside previous marketing tools for local and health-based merchants.

Financial Data Analytics for Insurance Profitability

2025

@ Saman Insurance

Creating Interactive Dashboards for Product Managers

This project focused on developing interactive dashboards for product managers, transforming complex actuarial and financial data into actionable visual insights. The goal was to empower managers to monitor key profitability metrics, identify trends in claims and premiums, and make data-driven decisions to optimize product performance and enhance the company's financial health.

This independent research initiative involved a deep dive into the emerging ecosystem of Decentralized Finance. It explored innovative use cases for blockchain-based financial instruments and analyzed the investment strategies of "smart money" investors to understand the practical applications and future potential of DeFi protocols beyond speculative trading.

This project involved researching and defining the core requirements for building Explainable AI (XAI) systems within a credit scoring context. The goal was to move beyond "black box" models by creating an architectural framework that ensures algorithmic decisions are transparent, interpretable, and compliant with regulatory standards, thereby building trust with both regulators and applicants.

Honors and Awards

Won	Gold medal for sharif university volleyball tournament Sharif University of Technology, Tehran, Iran, 2020
Ranked 6th	Among graduating class of industrial engineering at the end of bachelor's Sharif University of Technology, Tehran, Iran, 2020
Ranked 260th	Among approximately 260,000 participants in the university entrance exam Iran, 2016
Certified	National young mathematicians' association exam held by the University of Waterloo, Canada, 2013

References

Dr. Babak Haji. Assistant Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Shokrane K. Moghaddam. Assistant Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Nafiseh Sedghi. Assistant Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Erfan Hassannayebi. Assistant Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Shahram Shadrokh. Associate Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Mohammad Modarres Yazdi. Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran

Dr. Kourosh Eshghi. Professor,
Faculty of Industrial Engineering Department, Sharif University of Technology, Tehran, Iran